



ST BENEDICT'S

SUBJECT	PHYSICS FINAL EXAMINATION
GRADE	11
EXAMINER	Mr McCoy
MODERATOR	Mrs Govender
DATE	09 November 2023
MARKS	200
DURATION	3 hours

%

LEARNER NAME:

PLEASE READ THE FOLLOWING INSTRUCTIONS CAREFULLY

1. This question paper consists of 28 pages, including additional space. A Data Sheet has been provided. Read the questions carefully.
2. Answer **ALL** the questions on the question paper and hand it in at the end of the assessment. Remember to write your name and surname in the space provided above.
3. Use the data and formulae sheet whenever necessary.
4. Show your working in all calculations.
5. Units need not be included in the working of calculations, but appropriate units should be shown in the answer.
6. Answers must be expressed in decimal format, not left as proper fractions.
7. Where appropriate, express answers to TWO decimal places.
8. It is in your own interest to write legibly and to present your work neatly.
9. A blank page and axis is included at the end of the paper. If you run out of space for a question, use this page. **Clearly indicate the question number** of your answer should you use this extra space.

FOR OFFICE USE ONLY: MARKER TO ENTER MARKS

Question	1	2	3	4	5	6	7	8	9	10	Total
Mark											
Moderated Mark											
Question Total	20	20	29	18	20	21	30	10	20	13	200

Comment:

QUESTION 1 MULTIPLE CHOICE**20 MARKS**

Answer these questions on the multiple-choice answer sheet below. Make a cross (X) in the box corresponding to the letter that you consider to be correct.

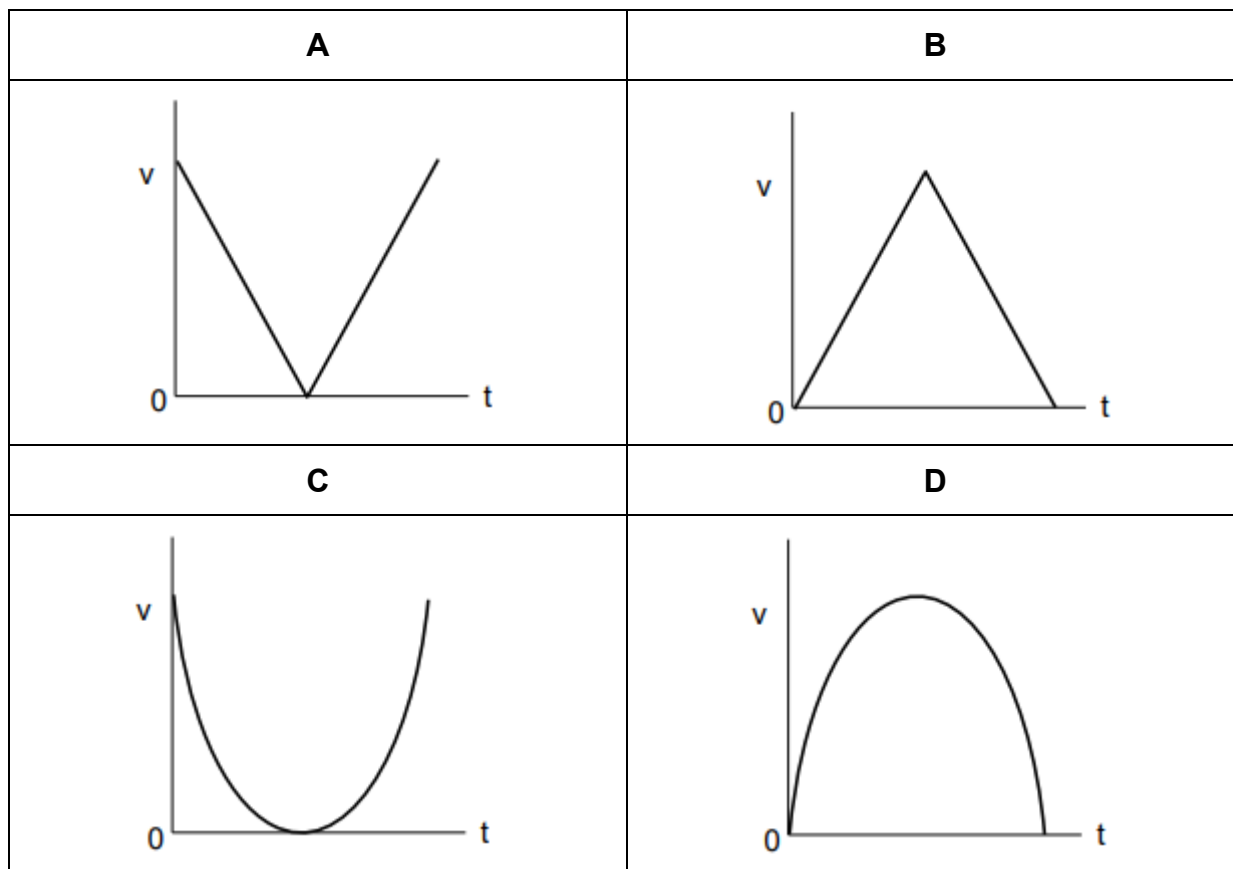
A	B	C	D
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Here the option C has been marked as an example.

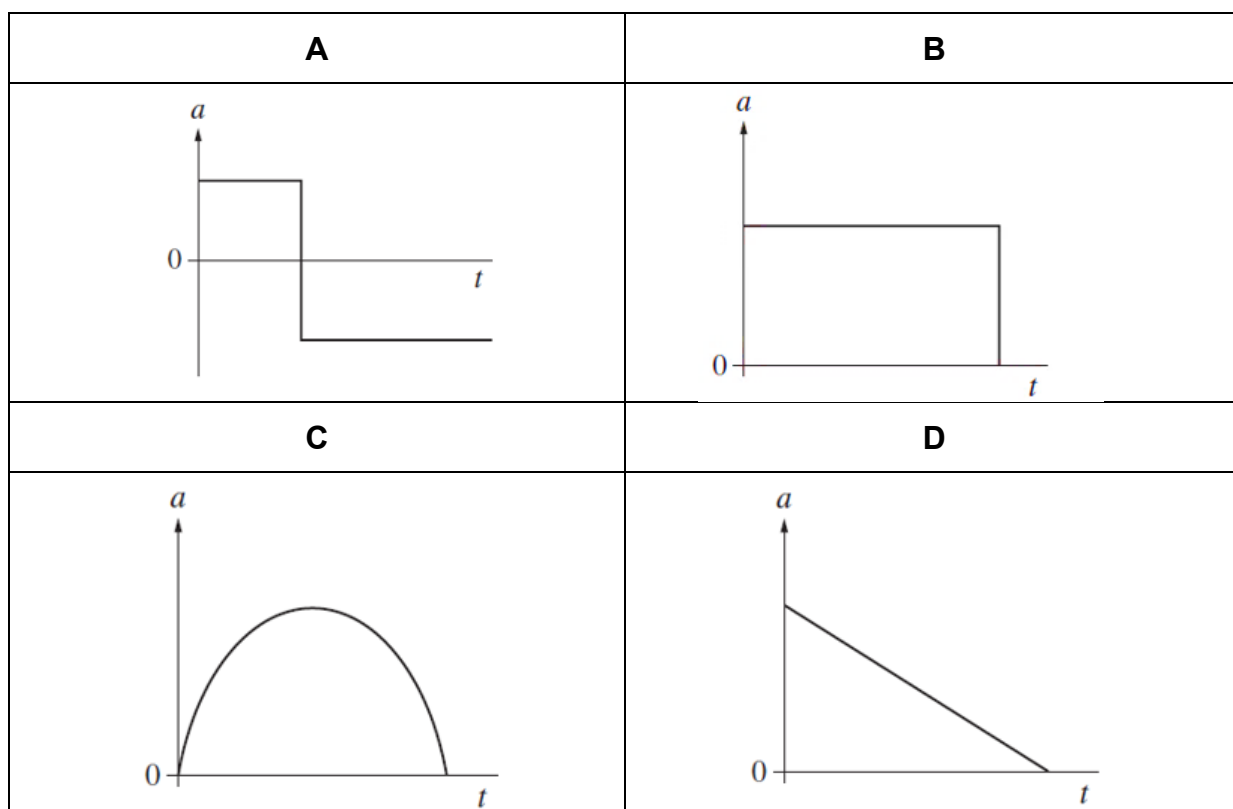
1.1	A	B	C	D
1.2	A	B	C	D
1.3	A	B	C	D
1.4	A	B	C	D
1.5	A	B	C	D
1.6	A	B	C	D
1.7	A	B	C	D
1.8	A	B	C	D
1.9	A	B	C	D
1.10	A	B	C	D

- 1.1 Which of the following is always true for an object moving in a straight line at constant speed?
- A No forces act on the object.
 - B No resultant force acts on the object.
 - C The kinetic energy of the object is zero.
 - D No work is being done on the object.
- 1.2 Two marbles, one of mass m and the other of mass $2m$, are dropped to the ground from the same height. Air resistance is negligible. Just before hitting the ground the kinetic energy of the lighter marble is E . This means that the kinetic energy of the heavier marble is:
- A E
 - B $2E$
 - C $4E$
 - D $0.5E$

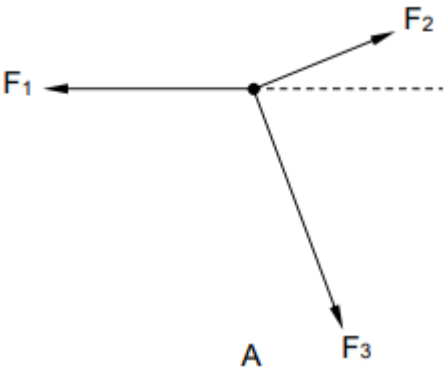
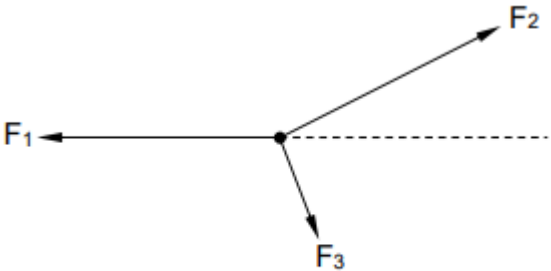
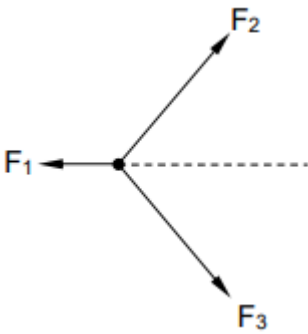
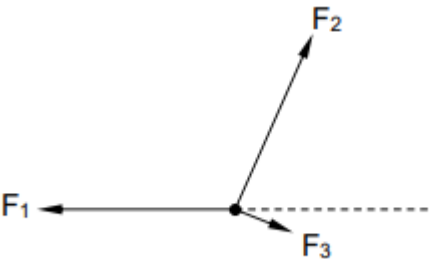
- 1.3 An object is thrown vertically upwards leaving the thrower's hand at time $t = 0$. Which graph shows how speed v varies with t as the object rises and falls?



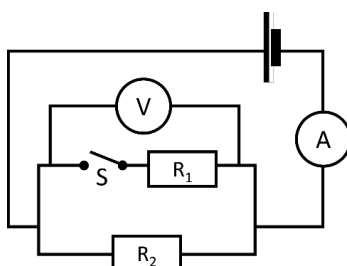
- 1.4 A stone is thrown horizontally from the top of a cliff and falls onto the beach below. Which acceleration–time graph best describes the motion of the stone?



- 1.5 Three forces, drawn to scale, act at a point. In which diagram is the point in equilibrium?

A	B
	
C	D
	

- 1.6 The cell, connecting wires and ammeter in the circuit represented below have negligible resistance. The switch is open and resistors R_1 and R_2 are connected in parallel.

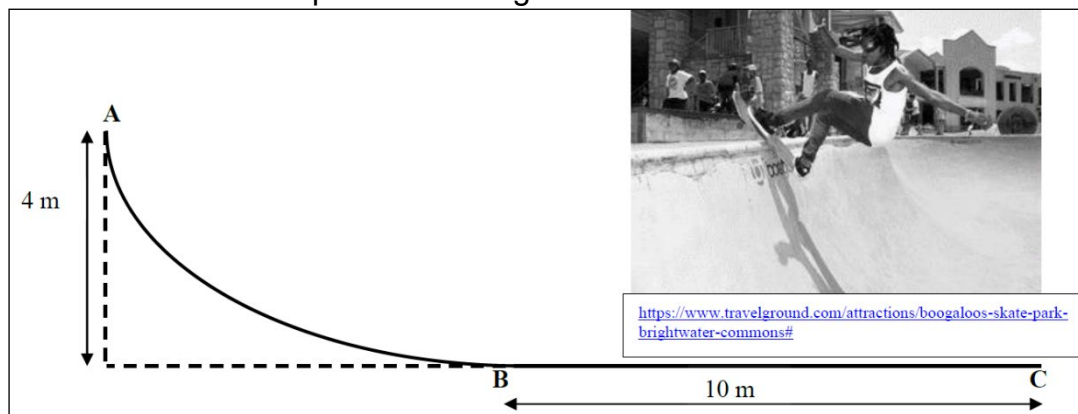


What happens to the voltmeter and ammeter readings when switch S is closed.

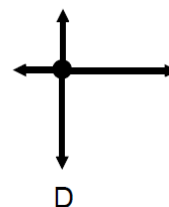
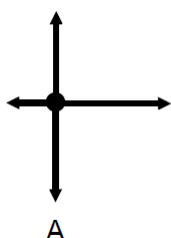
	Voltmeter reading	Ammeter reading
A	Remains the same	Increases
B	Increases	Remains the same
C	Remains the same	Decreases
D	Decreases	Remains the same

- 1.7 A skater on a skateboard is initially at rest at the top (position A) of a skate rink. She skates down the slope and reaches a speed of $7,5 \text{ m.s}^{-1}$ at position B.

The friction on the slope cannot be ignored.



Write down the letter that corresponds to the free body diagram of the forces acting on the skater at position B.



- 1.8 A boy and a girl sit on office chairs, which have wheels attached to them. The chairs are at rest on the horizontal floor. The floor is so smooth that we will assume that it is frictionless. The boy and his chair have greater mass than the girl and her chair.

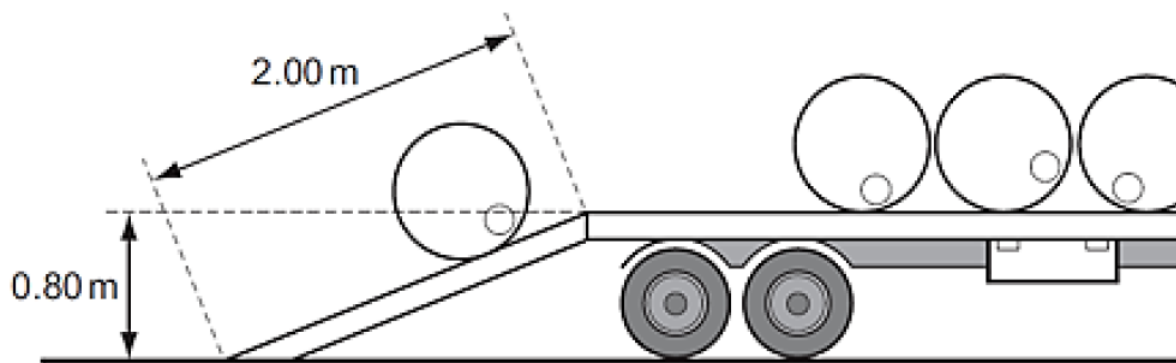


The boy and girl simultaneously push against each other.

Which of the following statements are both correct regarding their subsequent motion over the first 2 seconds?

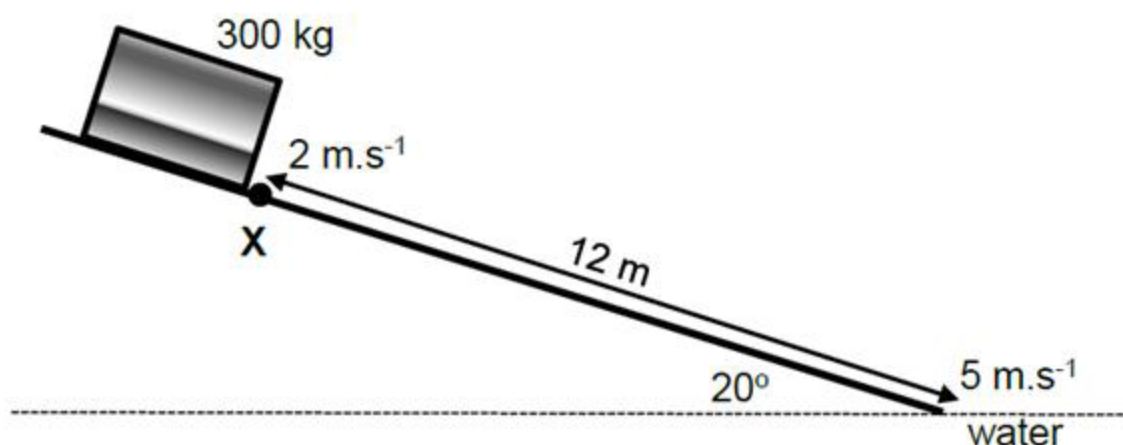
	Motion of person in first 2 s	Reason
A	The boy travels further than the girl	The force on the boy is greater
B	The boy travels further than the girl	His acceleration is greater
C	The girl travels further than the boy	The force on the girl is greater
D	The girl travels further than the boy	Her acceleration is greater

- 1.9 A workman rolls a barrel of weight 2000 N up a plank of length 2 m and on to a lorry. The back of the lorry is 0,8 m above the horizontal surface of the road.



What work is done on the barrel against gravity?

- A 1 000 J
 - B 1 600 J
 - C 2 500 J
 - D 4 000 J
- 1.10 A 300 kg box is placed on a slope at with an angle of 20° to the horizontal.



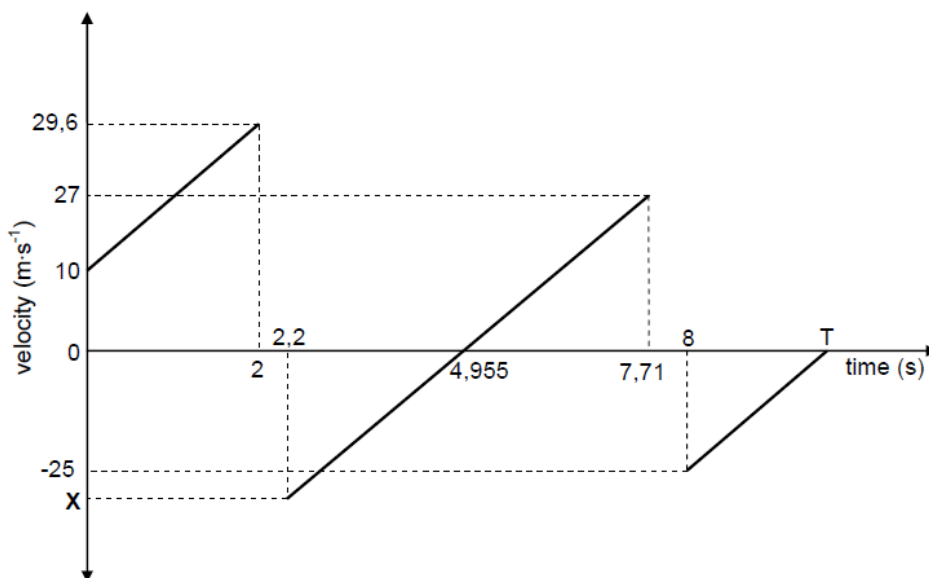
The angle of the slope is changed to 30° . How will this affect the following...

	Normal	Velocity of 5 m.s^{-1}	Net work done on box
A	Increase	Greater than	Decrease
B	Decrease	Greater than	Increase
C	Increase	Less than	Decrease
D	Decrease	No change	Increase

QUESTION 2 KINEMATICS: GRAPHS OF MOTION**20 MARKS**

A ball is thrown **vertically downwards** from the top of a building and bounces a few times as it hits the ground. The velocity-time graph below describes the motion of the ball from the time it is thrown to a certain time **T**.

Take downwards as the positive direction and the ground as zero reference. The graph is NOT drawn to scale. The effects of air friction are ignored.



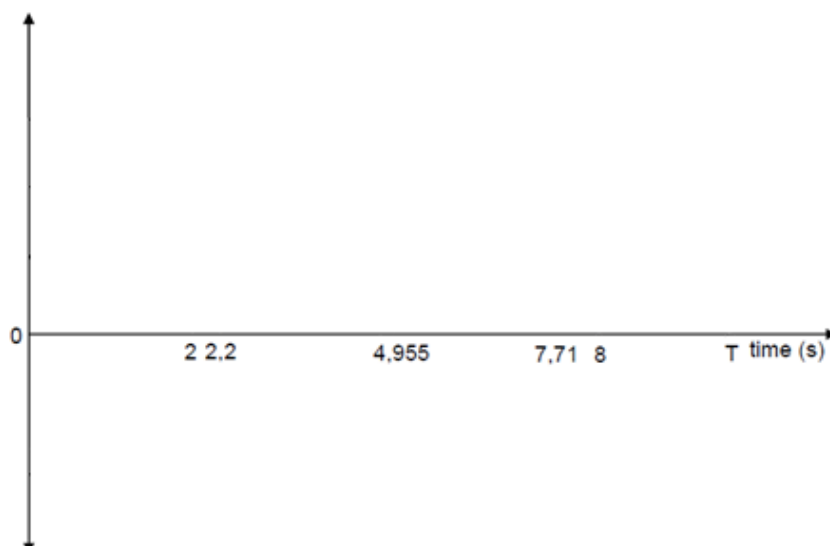
2.1 Define *velocity*. (2)

2.2 Write down the speed with which the ball is thrown downwards. (1)

2.3 ALL parts of the graph have the same gradient. Give a reason for this. (2)

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2.4 Sketch an acceleration versus time graph for the whole motion (2)



2.5 Calculate the:

2.5.1 height from which the ball is thrown (3)

2.5.2 time (T) shown on the graph (4)

2.6 Write down the:

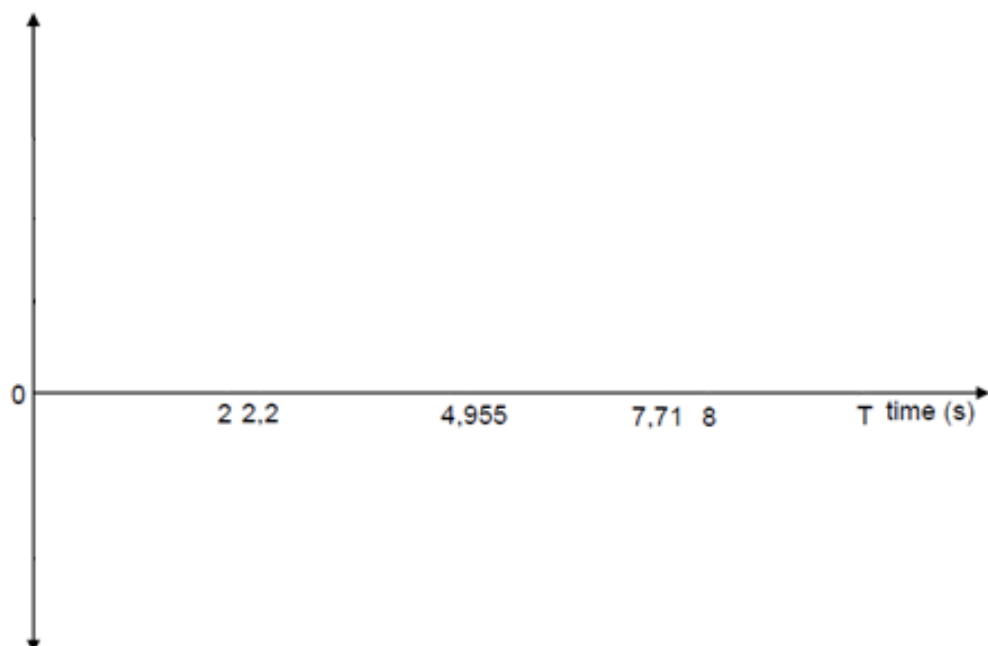
2.6.1 time that the ball is in contact with the ground at the first bounce (1)

2.6.2 time at which the ball reaches its maximum height after the first bounce (1)

2.6.3 value of X shown on the graph (1)

2.7 Sketch a displacement versus time graph of the motion of the ball up to 7,71s. Choose the ground as zero reference.

You do not need to indicate any displacement values on the vertical axis. (3)



QUESTION 3 KINEMATICS: HORIZONTAL MOTION**29 MARKS**

3.1 Niccolo runs a distance of 150 m in a time of 17 s.

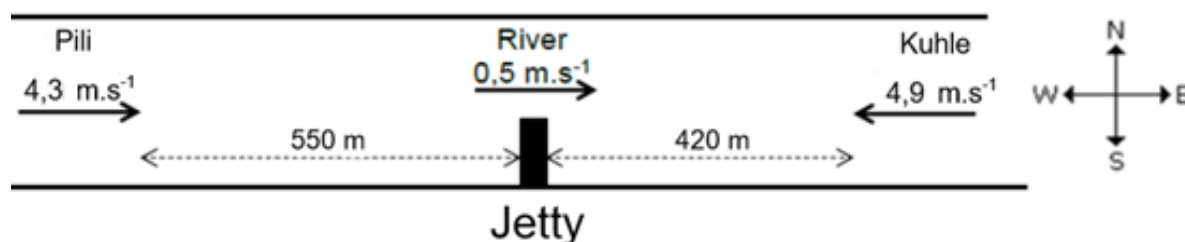
3.1.1 Define *speed*. (2)

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3.1.2 Calculate Niccolo's average speed. (3)

3.2 Pili and Kuhle are paddling in a river where the water is flowing at $0,5 \text{ m.s}^{-1}$ east.

- Kuhle who is 420 m east of the jetty paddles at $4,9 \text{ m.s}^{-1}$ west towards the jetty.
- Pili who is 550 m west of the jetty paddles at $4,3 \text{ m.s}^{-1}$ east towards the jetty.



3.2.1 Define *resultant vector*. (2)

3.2.2 Determine Kuhle's resultant velocity. (2)

3.2.3 By means of suitable calculations determine which paddler reaches the jetty first. Assume they both start paddling towards the jetty at the same time. (5)

3.3 A car is accelerated uniformly from rest at $4,2 \text{ m.s}^{-2}$ north. At the instant the car starts to accelerate there is a truck moving at a constant speed of $7,8 \text{ m.s}^{-1}$ north in the same lane a distance “**x**” metres ahead of the car. It takes the car exactly 8 s to catch up to the truck.

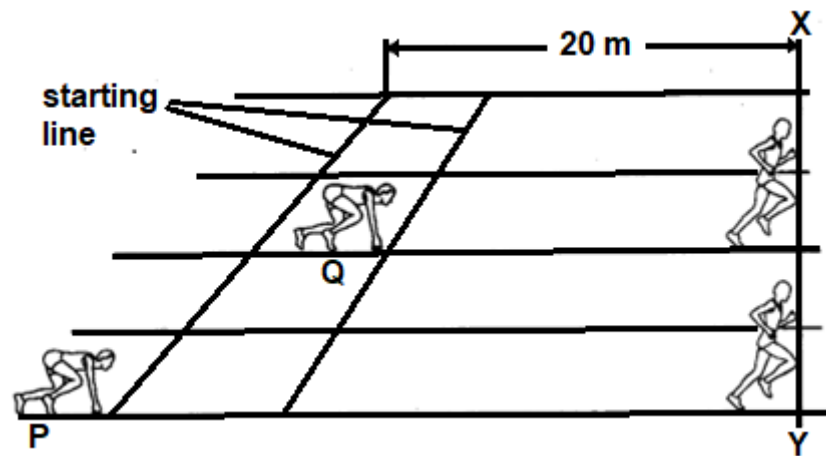
3.3.1 Calculate the speed reached by the car in 8 s. (3)

3.3.2 Sketch a velocity versus time graph for the motion of the car and truck. (3)

3.3.3 Calculate the original distance “**x**” between the car and the truck. (5)

- 3.4 In a "handicap" sprint race, sprinter P and Q both start the race at the same time but from different starting lines on the track.

The handicapping is such that both sprinters reach the line XY, as shown below, at the same time.

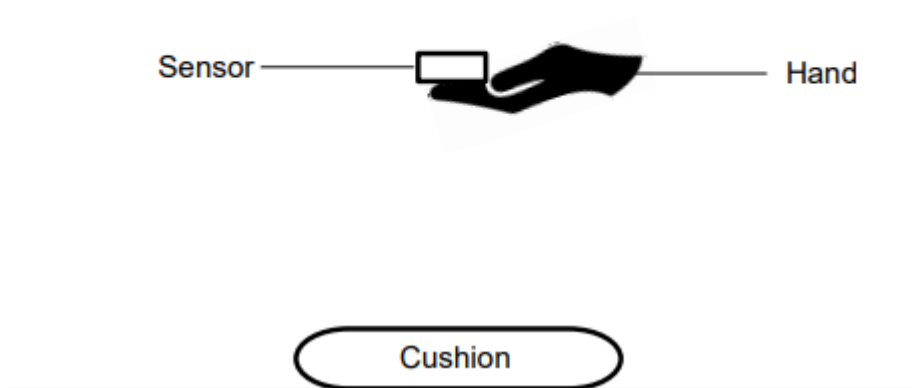


Sprinter P has a constant acceleration of P from the start line to the line XY
Sprinter Q has a constant acceleration of Q from the start line to the line XY

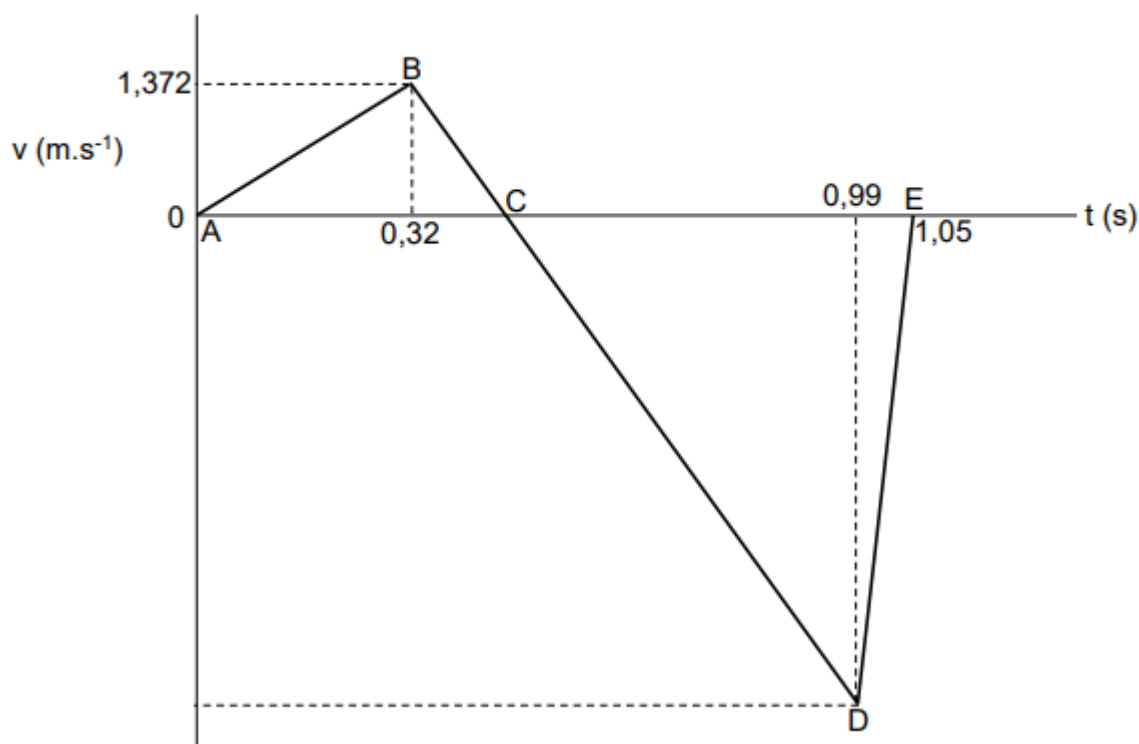
Find the speed of **Sprinter Q** at line XY, in terms of acceleration P and Q (4)

QUESTION 4 KINEMATICS: VERTICAL MOTION**18 MARKS**

A student throws a 250 g sensor vertically upwards from rest with a constant force and then the sensor eventually lands on a cushion below his hand. Ignore air resistance.



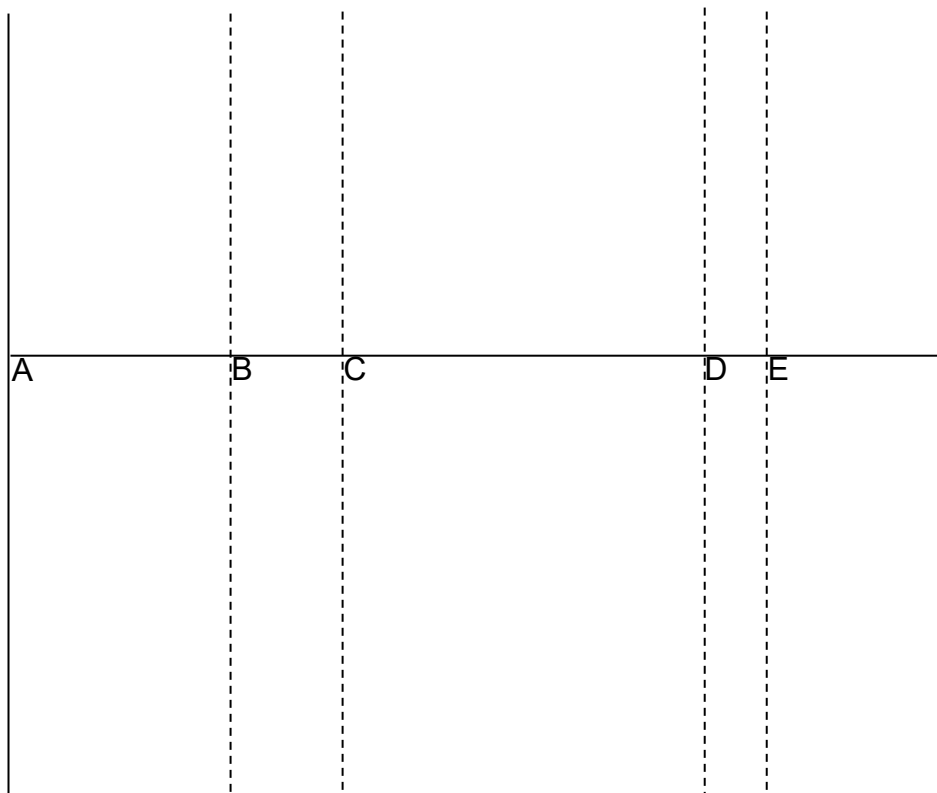
The sensor collects data and the velocity versus time graph of the entire motion of the sensor is shown below.



- 4.1 What is the direction of the acceleration of the sensor during the time interval $t = 0$ to $t = 0.32 \text{ s}$? (1)

- 4.2 During which time interval is the sensor in free fall? (1)

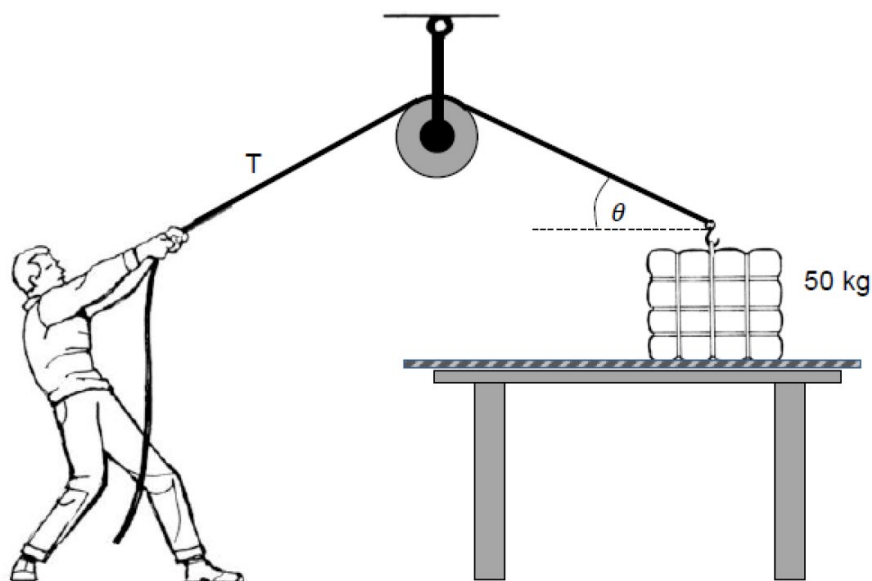
- 4.3 Use the graph to calculate the distance over which the hand was in contact with the sensor. (3)
- 4.4 Calculate the total time taken for the sensor to reach its maximum height. (3)
- 4.5 Calculate the height of the sensor above the cushion at $t = 0$. (5)
- 4.6 Draw a neat position versus time graph for the motion of the sensor for the time interval $t = 0$ to $t = 1,05$ s. Take the initial position of the sensor as the reference point. No numbers are required on the axes. (5)



QUESTION 5 FORCES ON ONE BODY**19 MARKS**

The man is unable to lift the heavy parcel because:

- He's applying a constant pulling force.
- The mass of the parcel is 50 kg.
- There's a light inextensible rope passing over a frictionless pulley.
- The coefficient of static friction between the parcel and the rough table is 0,34.
- The maximum static frictional force is 120 N.
- The mass of the rope is ignored.



5.1 Draw a free-body diagram showing all the forces exerted on the parcel. (4)

5.2 Define *frictional force due to a surface*. (2)

5.3 Calculate the normal on the parcel when the static friction is at a maximum. (2)

5.4 When the static friction is at a maximum, show that the magnitude of the vertical component of the tension force in the rope is 137,06 N. (3)

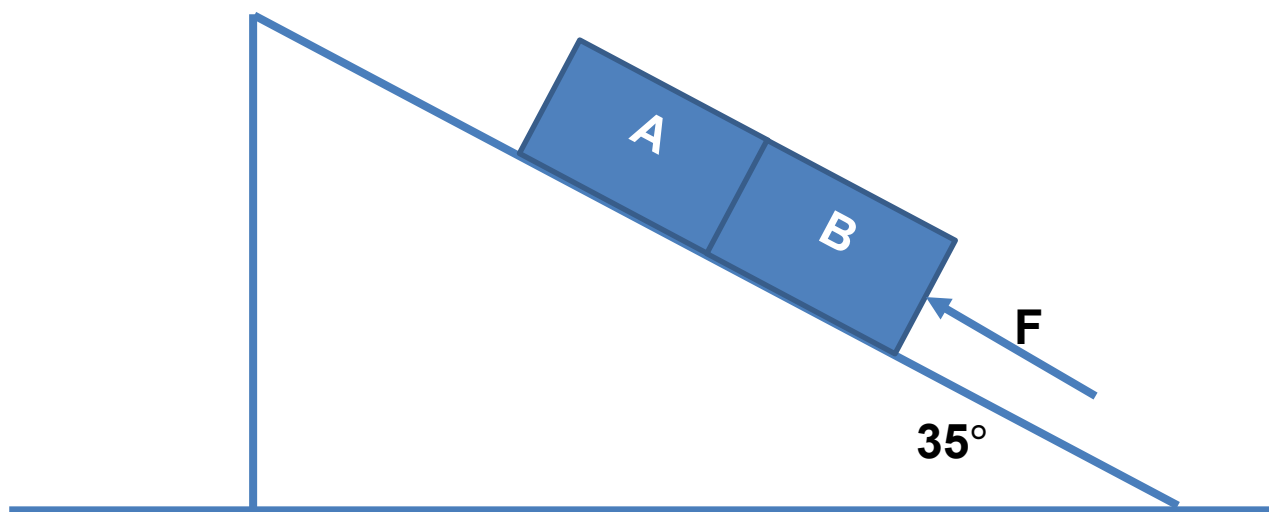
5.5 Hence, calculate the angle (θ) that the rope forms with the horizontal. (3)

5.6 Calculate the magnitude of the tension in the rope (T). (2)

5.7 The man pulls the rope quickly and it breaks.
Use Newton's laws to explain why. (3)

QUESTION 6 FORCES ON TWO BODIES**21 MARKS**

A copper block A and a steel block B, each of mass 0,4 kg are only just held at rest by a force F acting up the slope as shown in the sketch below. The coefficient of static friction between block A on the slope is 0,52.



6.1 Define *weight*. (2)

6.2 Draw a labelled free body diagram showing all the forces acting **on block A** while at rest. (4)

6.3 Calculate the maximum frictional force acting on block A. (5)

The maximum frictional force on block B is 1,3 times the frictional force on block A.

6.4 Explain how block B can have a larger maximum frictional force than block A. (2)

The force F is removed and the blocks accelerate down the slope. Block A experiences a frictional force of 0,9 N and block B experiences a frictional force of 1,17 N.

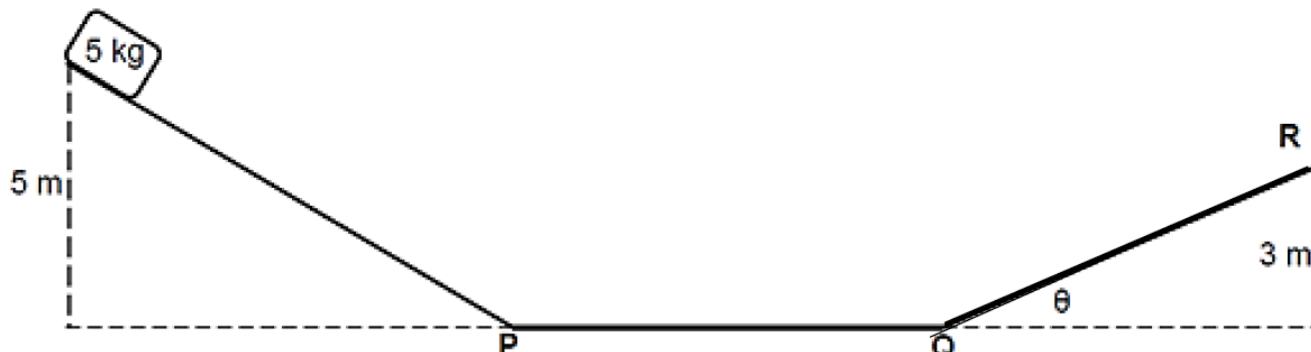
6.5 Why is the frictional force on block A now less than that calculated in question 6.3? (2)

The acceleration of the blocks is measured to be $3,03 \text{ m.s}^{-2}$ down the slope.

6.6 Calculate the force exerted by block B on block A. (6)

QUESTION 7 WORK, ENERGY AND POWER**30 MARKS**

- 7.1 A 5 kg block is released from rest from a height of 5 m and slides down a **frictionless** incline to point P as shown in the diagram below. It then moves along a **frictionless** horizontal path PQ and finally moves up a second **rough** inclined plane. It comes to rest at point R which is 3 m above the horizontal.



- 7.1.1 State the *principle of conservation of mechanical energy*. (2)

- 7.1.2 Use the principle of conservation of mechanical energy to calculate the speed of the block at point P. (4)

- 7.1.3 Explain by identifying the relevant law of physics, why the kinetic energy at point Q is the same as that at point P. (3)

The frictional force between the surface and the block from Q to R is 18 N.

- 7.1.4 Which force acting on the block from Q to R does no work on the block? (2)

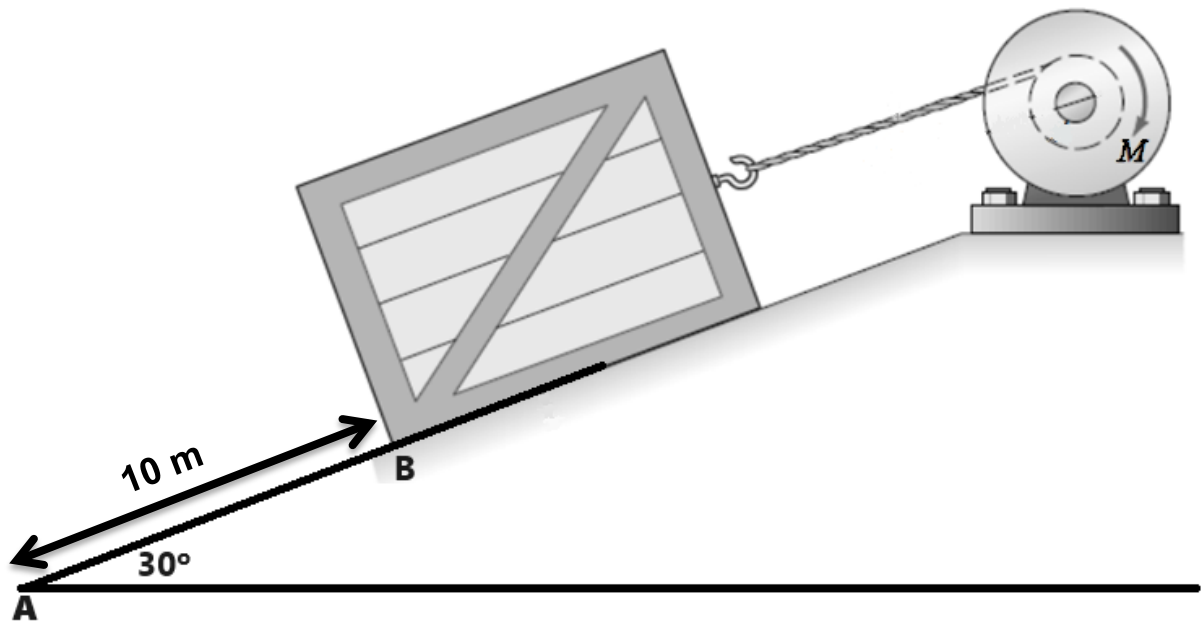
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7.1.5 Calculate the work done against friction from Q to R. (4)

7.1.6 State the *work-energy theorem*. (2)

7.1.7 Hence, use the work-energy theorem calculate the angle (θ) to the horizontal of the slope QR. (5)

- 7.2 You have a winch with a 210 W motor pulling a 20 kg crate up a 30° slope from point A to B over a 10 m distance. The crate experiences friction, takes 8 seconds to complete the motion, and the tension in the cable is 150 N.



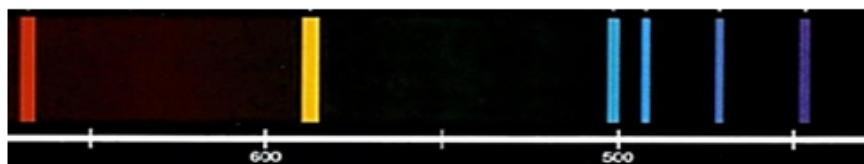
- 7.2.1 Define *power*. (2)

- 7.2.2 Calculate the power output of the electric motor moving the crate from point A to point B. (3)

- 7.2.3 Calculate the efficiency of the electric motor. (3)

QUESTION 8 PHOTONS AND ELECTRONS**10 MARKS**

- 8.1 Each element emits its own unique line emission spectrum when high voltage is applied across a low-pressure vapour of its atoms enclosed in a glass tube. Below is the line emission spectrum of an element.



- 8.1.1 Calculate the energy of a photon of blue light with a wavelength of 492,1 nm. (3)

- 8.1.2 What causes a line spectrum to be emitted? (3)

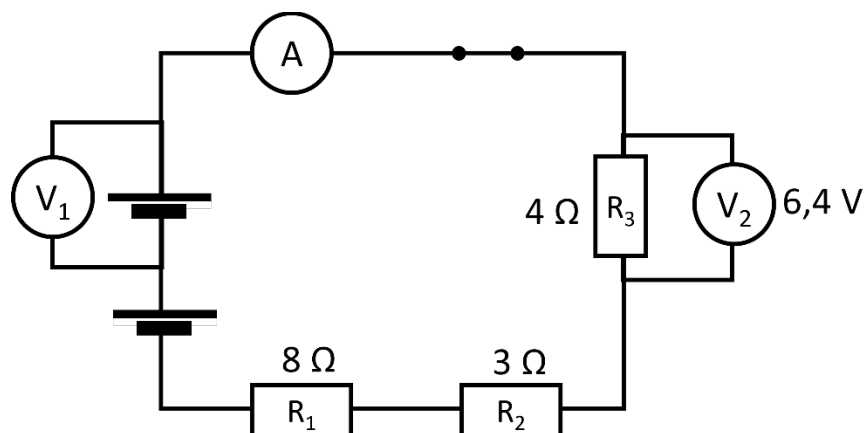
- 8.1.3 How many energy levels would be required to produce the line emission spectrum above? (2)

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- 8.2 Explain why each element emits a unique line emission spectrum. (2)

QUESTION 9 ELECTRIC CIRCUITS**20 MARKS**

In the circuit shown below. Two identical cells are connected in series circuit containing three resistors, a switch, and an ammeter. A voltmeter (V_2) is connected over the $4\ \Omega$ resistor and displays a reading of $6,4\ \text{V}$ when the switch is closed.



9.1 State *Ohm's Law*. (2)

9.2 Determine the reading on the ammeter. (2)

9.3 How many Coulombs of charge move through the $8\ \Omega$ resistor per second? (2)

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9.4 Determine the amount of work done by R_1 in 10 minutes. (4)

9.5 Calculate the reading on voltmeter V_1 . (4)

9.6 Define *resistance*. (2)

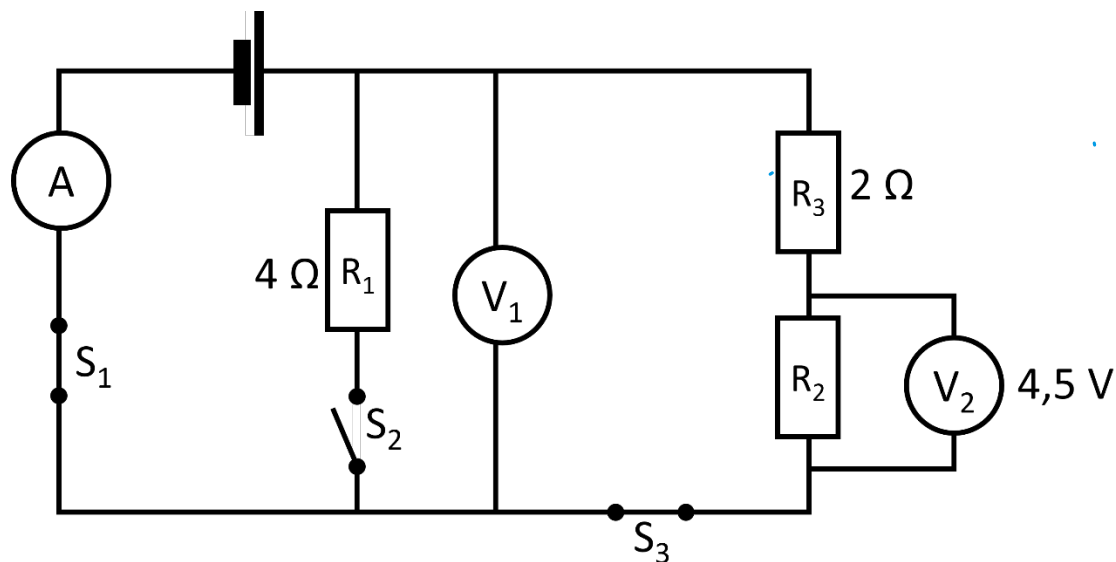
The $3\ \Omega$ resistor (R_2) is removed and the circuit is reconnected.

9.7 How will the current in the circuit change?
Circle: **INCREASE, DECREASE or REMAINS THE SAME.** (1)

9.8 Explain your answer to QUESTION 9.7 (3)

QUESTION 10 ELECTRIC CIRCUITS**13 MARKS**

In the following circuit the cell has an unknown emf. Switches S_1 and S_3 are initially closed while S_2 is open. While the switches are in this arrangement ammeter A has a reading of 0,75 A and voltmeter V_2 displays a reading of 4,5 V. The resistance of R_1 is $4\ \Omega$, and R_3 is $2\ \Omega$, the resistance of R_2 is unknown.



10.1 Determine the resistance of R_2 . (2)

10.2 How much power does the cell supply to the circuit? (3)

Switch S_2 is now closed

10.3 What is the reading on V_1 . (1)

10.4 Determine the effective resistance of the circuit.

(4)

Switch S_2 remains closed and Switch S_3 is opened.

10.5 How will the reading on voltmeter V_1 change **INCREASE**, **DECREASE** or **REMAIN THE SAME**? Circle the correct option.

(1)

10.6 Explain your answer to QUESTION 10.5.

(2)

END OF ASSESSMENT

Acknowledgments:

ST ANNE'S DIOCESAN COLLEGE 2019

REDHILL HIGH SCHOOL 2019

ADDITIONAL SPACE

REMEMBER TO CLEARLY INDICATE AT THE QUESTION THAT YOU USED THE
ADDITIONAL SPACE TO ENSURE THAT ALL ANSWERS ARE MARKED

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[illegible]

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